Statistical Analysis for Tip Type Data
Completely Randomized Block Design

The experimental design used for the Tip Type data, presented in Lecture Notes no 7, is a completely randomized block design, using the following model

$$Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij},$$

where $1 \leq i \leq a$ and $1 \leq j \leq b$. In our experiment a = 4 and b = 4. In the model above, the effect of the ith tip type is a fixed effect. The effect of the jth block, the jth metal coupon, is a random effect. In this experiment, based on the writeup in Lecture Notes no 8, we are interested in testing the following null hypothesis:

$$H_0: \tau_1 = \tau_2 = \tau_3 = \tau_4 = 0, \qquad (1)$$

Which is equivalent to testing

$$H_0: E(\mu_1)=E(\mu_2)=E(\mu_3)=E(\mu_4),$$

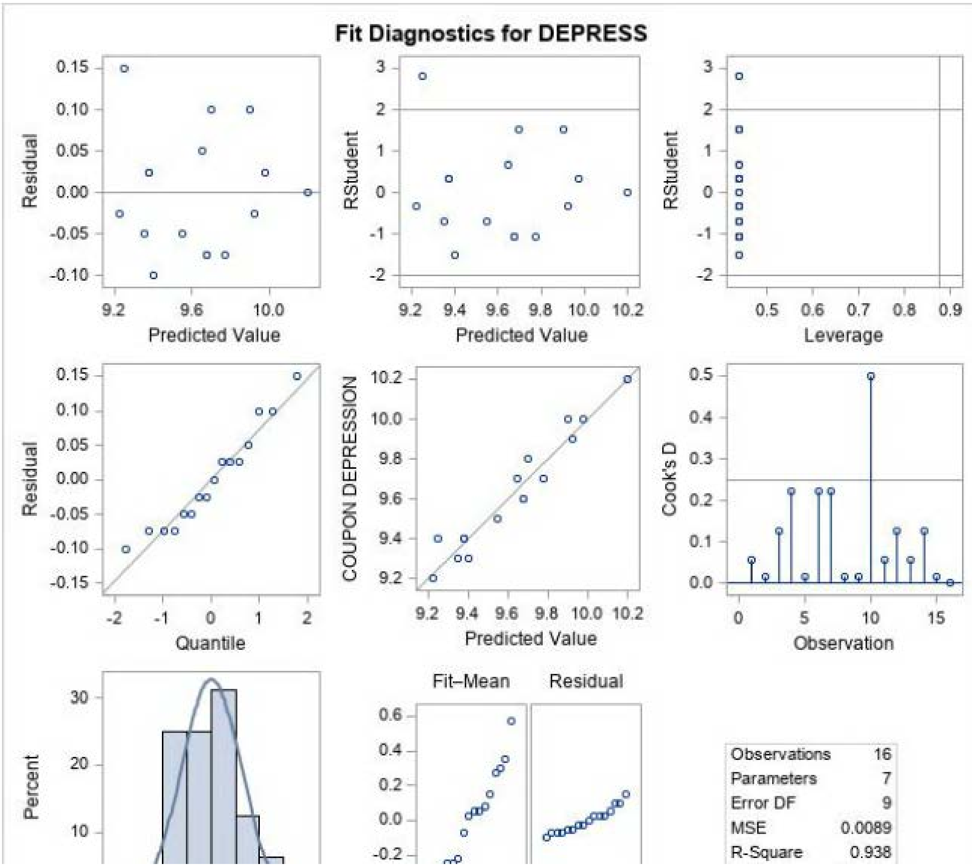
if we define $\mu_i = \mu + \tau_i$, where $1 \leq i \leq a$.

The SAS file and output for the Tip Type data is posted on Husky CT cite for the course. As we mentioned in class, all the underlying assumptions for the model above are valid. For example, the normality assumption for the random errors is confirmed based on the normal probability plot (Residual vs. Quantile plot) in the Fit Diagnostics pane plots presented below, and the fact that the p-value of the W statistic is equal to .3438, which is larger than .1.

Tests for Normality

Test	Statistic		p Value	
Shapiro-Wilk	W	0.939575	Pr < W	0.3438
Kolmogorov-Smirnov	D	0.133947	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.055036	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.375139	Pr > A-Sq	>0.2500

The assumption of linearity of the model and constant variance of random errors is validated via the plot of Residuals vs. Predicted values, as the points in that plot are scattered randomly about the 0 line and do not show any patterns.

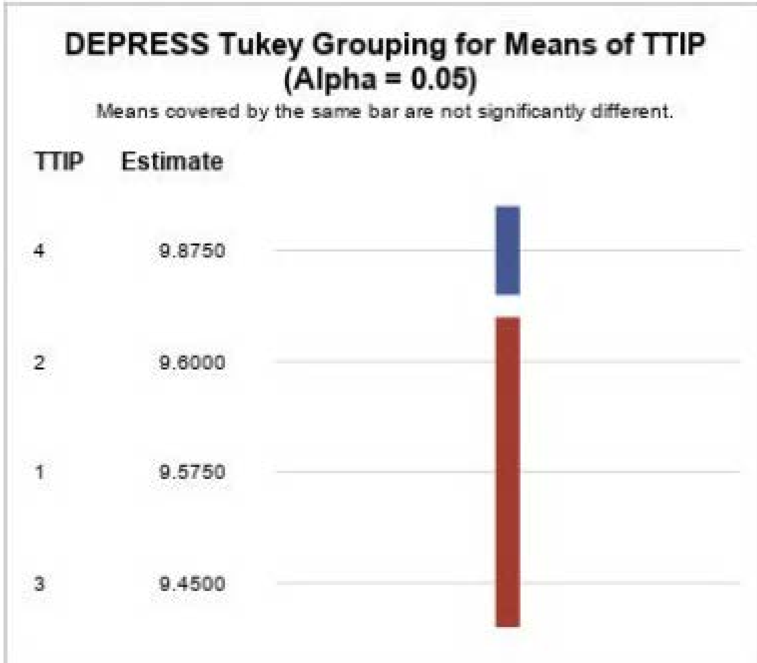


Therefore, from the SAS output it follows for the table below

Source	DF	Type III SS	Mean Square	F Value	Pr > F
TTIP	3	0.38500000	0.12833333	14.44	0.0009
COUPON	3	0.82500000	0.27500000	30.94	<.0001

That the F statistic for testing hypothesis (1), has an observed value of 14.44 and a p-value of .0009. Therefore, we can reject the null hypothesis in (1) at the .001 level.

This means that the four Tip Types do not have the same strength. Based on the Tukey multiple comparisons graphical display in the SAS output:



Tip Type no 4, that has a sample mean of 9.875, is significantly stronger than the other three Tip Types. The other three Tip Types are not significantly different from each other.